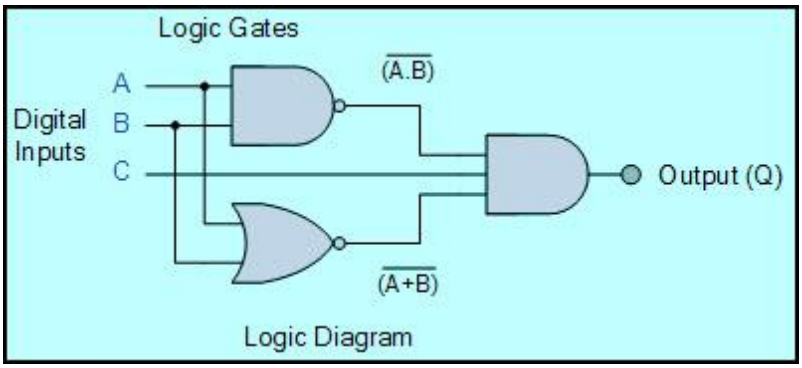
Example of relay wiring

Implement using switches and relays the following combinational circuit is



And the output is

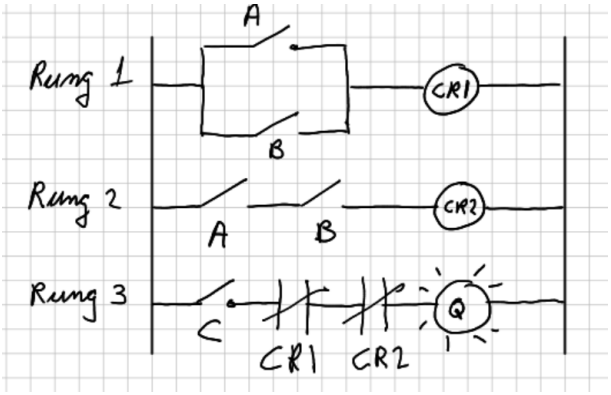
$$Q = \overline{(AB)}\,\overline{(A+B)}C$$

This output can be simplified but we will leave that for later. First, we are going to implement the most complicated solution (without simplification) using switches and relays. Remember that A, B and C are coming from switches.

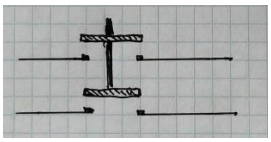
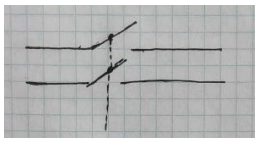
Using the switches directly

One way to do this is to use the switches directly to implement the relations:

$$\overline{(AB)}$$
$$\overline{(A+B)}$$

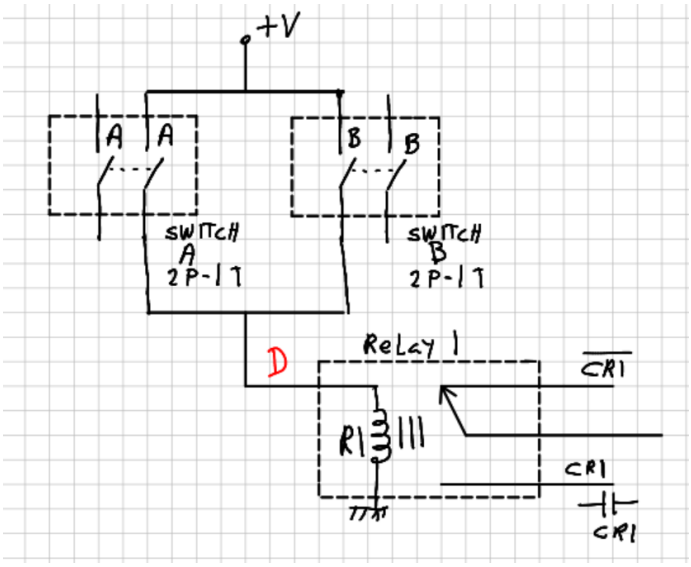


Since A and B are used in rung 1 and rung 2, we need them to be switches with 2 poles and one throw. One pole is needed for the circuit in rung 1 and the other pole is needed for the circuit in rung 2. Two simple drawings of this type of switches are next:

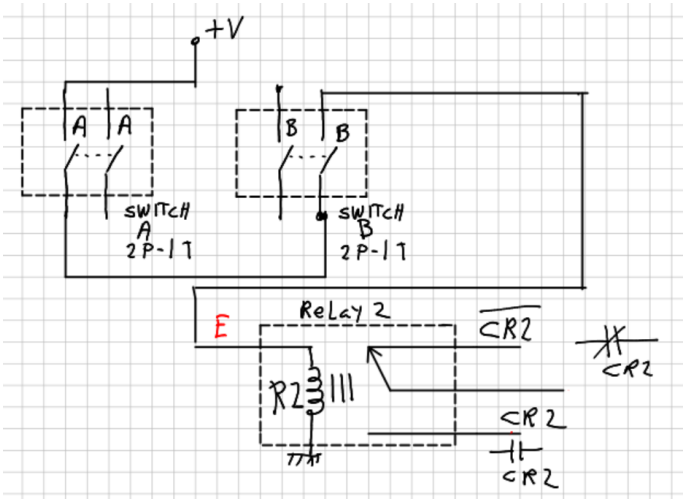



CR1 and CR2 are control relays needed as intermediate outputs, these relays must present a switch with one pole and two throws, one throw is normally open, and the other throw normally closed, as typically done. We want the normally closed terminals for rung 3.

The circuit/wiring for rung 1 is:

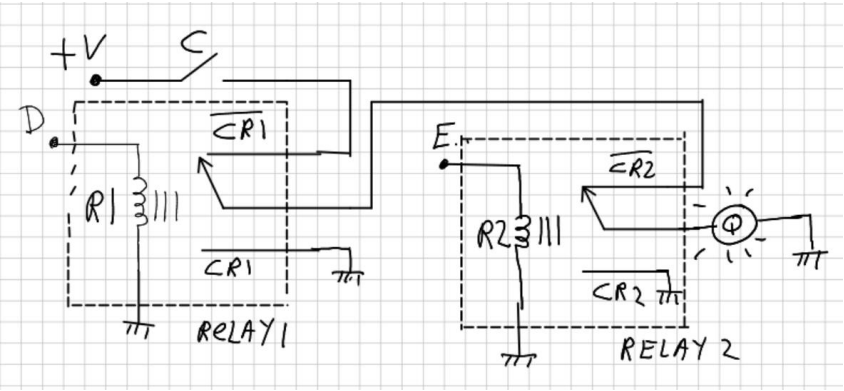


The wiring for the second rung is:



Notice that the wiring of rung 1 is not drawn in the drawing of the second rung to simplify the circuit; but keep in mind that both rungs are wired simultaneously.

The wiring for the third rung is (not including the first two rungs since it would be too messy):



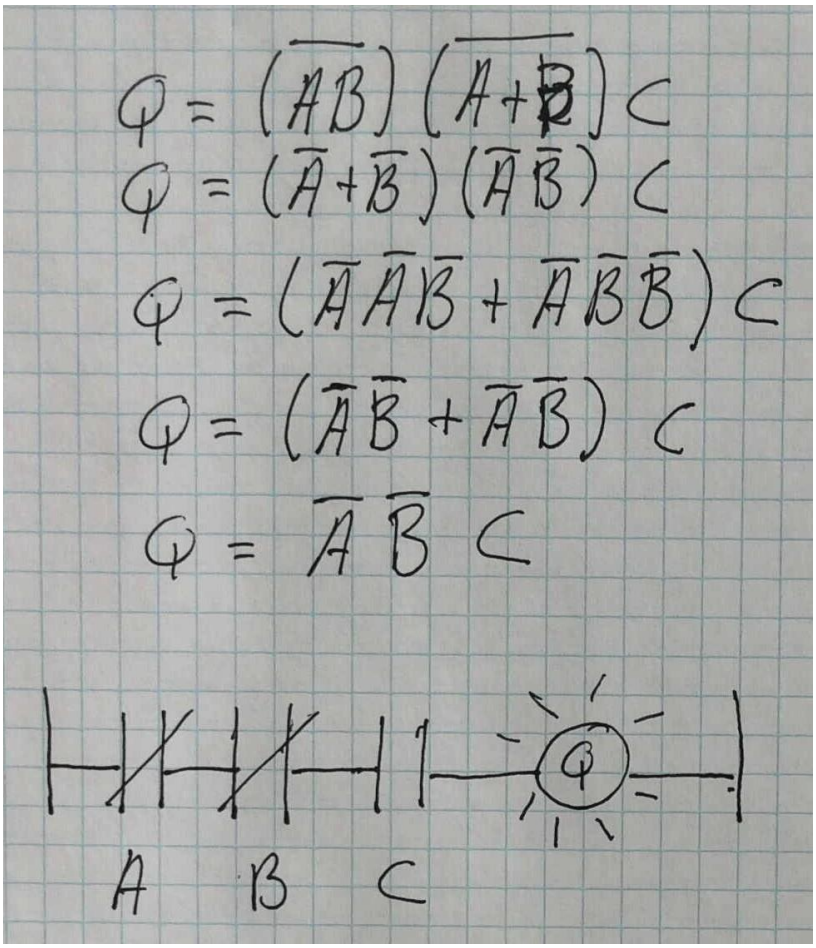
The Problem with relay logic wiring and logic simplification

This ladder is more complicated than it should be. When using relays (no PLCs) the wiring becomes very messy. I have to separate the wiring of each rung for clarity; otherwise, one single drawing showing the three rungs would be too intricate. Another big disadvantage of relay logic vs. PLCs is that re-wiring due to a design change or simply to mistakes is very costly because the physical wiring has to change; with PLCs, to fix a mistake is as simple as redrawing a rung in the ladder diagram (more about this later).

We can simplify the ladder diagram using De Morgan's law:

$$\overline{AB} = \bar{A} + \bar{B}$$
$$\overline{(A + B)} = \bar{A}\,\bar{B}$$

Therefore,



Using Morgan again, we can see that the result for Q above can be written as:

$$Q = C\bar{B}\bar{A} = C\overline{(A+B)}$$

Draw the RLD wirings for the two simplified versions above, the one for $C\bar{B}\bar{A}$ and the one for $C\overline{(A+B)}$. Which one do you prefer and why?

A problem doing logic using relays is that the negation of a signal has to be built in the hardware. This is why relays come with a normally closed connection (the negation) and a normally open connection (the normal signal) to guarantee we have CR and $\overline{\text{CR}}$ available all the time. With PLC, the negation of the signal is just a trivial step done in the software. Things changed for better with PLC.